

BRIAN E. PINEDA

US Citizen | New Brunswick, NJ

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Education

Rutgers University

Sep 2023 – May 2029 (Expected)

B.S. in Computer Science and Electrical Engineering — GPA: 3.8/4.00 — Dean's List

New Brunswick, NJ

Relevant Coursework: Artificial Intelligence, Data Structures, Design & Analysis of Algorithms, Data Management, Programming Languages, System Programming, Computer Architecture, Discrete Structures

Experience

Software Engineering Intern

May 2026 – Present

361Firm

Remote

- Audited a **40,000+ line React/Next.js** platform serving **25,000+ investors** and **900+ deals** by producing structured documentation of functional zones, resulting in an accessible internal reference and identified targets for modularization.
- Cut interactive **SVG world-map** marker rebuilds from **8 per pan to 0** by applying **React 19 memoization** across **1,140 mapped entities**, pinned by a render-count regression test to prevent silent regressions.
- Shipped **3 merged full-stack features** — a meeting-transcript insights pipeline, an idempotent event check-in dedupe library, and **WorldMap** UX fixes — on **Vercel-deployed Next.js 15**, adding **700+ lines** across UI, lib, and tests.
- Stood up the **first test coverage** on a **15,000-line** untested monolith by introducing **5 Vitest suites** totaling **600+ lines**, all running inside an enforced CI gate of typecheck, lint, build, and a **2,541-test suite**.

Projects

GenCost — Gen-AI Cost Metering & Routing Engine | *Python, FastAPI, Next.js, Supabase*

April 2026

- Built a **metering proxy** and **smart router** (**FastAPI** backend, **Next.js 15** + **Supabase** dashboard, **WebSocket**) that intercepts every generative-AI API call across **6 providers** spanning LLM, voice, image, and video, logging real **cost, latency, and lineage** in real time.
- Engineered a **cost-aware router** that sends each request to the cheapest model meeting a **Draft/Standard/Premium** quality tier, with **fallback chains** and **per-provider circuit breakers**, and covered the **5,000-line backend** with **39 unit tests**.

PC Forecaster — ML Price-Forecasting Web App | *Python, scikit-learn, Flask, SQLite, Pandas*

Jan – May 2026

- Built a PC-component **price-forecasting web app** (**Flask** + **scikit-learn**, **SQLite**, deployed on **Vercel**) that fits **Random Forest** models to project **12 months** of prices and emit buy-or-wait calls; out-of-time backtests on **559–1,689 held-out prices** reached an R^2 of **0.85–0.89** (MAPE as low as **15.7%**), with per-category **tier classifiers** at **83% GPU accuracy**.
- Engineered a **5-stage Python pipeline** that cleaned **9,452 raw records** into **6,078 unique components** via IQR filtering and spec parsing, matched **2,107 parts** across 2023–2025 snapshots, and synthesized a **61,103-row** monthly price series with **9 encoded market events**.

FlexMore — On-Device Pose-Estimation Form Coach | *TypeScript, React, Vite, MediaPipe*

June 2026 – Present

- Built a hands-free **webcam workout coach** that runs **MediaPipe** pose estimation **fully on-device** (**33 landmarks**, no video leaves the machine), counting reps and checking range of motion in real time with a **rep state machine**.
- Architected a **framework-free, deterministic analysis core** (one-euro smoothing → angle computation → rep state machine → structured set summary) so the vision layer measures and the LLM only phrases coaching, and covered it with **29 Vitest unit tests** runnable in bare Node.

Technical Skills

AI & ML: agentic AI design, prompt engineering, LLM API integration, scikit-learn, Random Forest models, MediaPipe

Languages: Python, TypeScript, JavaScript, Java, SQL, C, C++, HTML/CSS

Web & Frameworks: React, Next.js, FastAPI, Flask, Node.js, Vite, Tailwind, REST APIs

Data & Tools: Pandas, SQLite, Firebase (Auth/Firestore/Storage), Supabase, Git/GitHub, Vitest, AWS, Vercel

Leadership / Extracurricular

Public Relations Chair

September 2025 – May 2026

Mexican American Student Association (MASA)

Rutgers University, New Brunswick, NJ

- Design 2–3 promotional pieces weekly using coordinated messaging with the executive board, driving stronger event engagement and visibility across the organization.
- Lead the organization's social media content strategy and digital presence, ensuring consistent communication and brand alignment.